CSE374: Machine Learning Classification Project

Final Project Report

Dataset: Credit Card Default (Taiwan)

Models: Logistic Regression | Decision Tree | k-Nearest Neighbours

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Course: CSE374 — Machine Learning and Pattern Recognition

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Section 1: Problem Framing & Data Understanding

1.1 Problem Description

Banks and financial institutions need to assess the credit risk of their customers before extending or renewing credit lines. The core question is: given a customer's demographic profile, credit history, and recent payment behaviour, can we predict whether they will default on their next monthly payment? Accurately identifying likely defaulters allows the bank to take preventive action before a loss occurs. This study applies supervised machine learning classification to the Credit Card Default dataset from Taiwan to build and compare predictive models for this task.

1.2 Target Variable

The target variable is default.payment.next.month : a binary indicator where 1 means the customer defaulted and 0 means they did not. In the notebook it was renamed to "default" for readability. The prediction task is therefore a binary classification problem with two possible outcomes per customer.

1.3 Classification Type

This is a binary classification problem. The model outputs one of two outcomes: default (1) or no default (0). Binary classification enables the use of standard evaluation metrics including Accuracy, F1-Score, Confusion Matrix, and ROC/AUC, all of which are reported in this study.

1.4 Anticipated Challenges

Challenge	Description	Mitigation Strategy
Class Imbalance	~22% default rate	class_weight="balanced"; report F1 and ROC-AUC as primary metrics
Feature Correlation	Six monthly BILL_AMT columns highly correlated (0.7-0.9)	Noted as LR limitation; discussed in error analysis
Mixed Feature Types	Categorical integers (SEX, EDUCATION, MARRIAGE) + continuous amounts	Excluded categoricals from StandardScaler
Noisy Categoricals	Undocumented values in EDUCATION (0,5,6) and MARRIAGE (0)	Merged into existing "other" categories during cleaning

Section 2: Data Acquisition & Documentation

2.1 Dataset Source

It was originally collected by researchers at National Cheng Kung University, Taiwan, and covers credit card clients from April to September 2005. It has been widely used in academic literature on credit risk modelling.

https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients

2.2 Dataset Overview

Property	Detail
Total Instances	30,000
Input Features	23
Target Column	default.payment.next.month (renamed to "default")
File Format	XLS (Excel 97-2003 — requires xlrd library)
Missing Values	None (confirmed via df.info())
Class Balance	22.1% default (class 1) / 77.9% no default (class 0)
Time Period	April–September 2005 (6 months of payment history)

Section 4: Data Cleaning & Preprocessing

All preprocessing was performed in a deliberate sequence: inspection → cleaning → splitting → scaling. This ordering is critical :scaling is applied after splitting to prevent data leakage from validation/test statistics into the training scaler.

4.1 Column Renaming and ID Removal

The target column was renamed from "default payment next month" to "default". The ID column was dropped as it is a sequential row number with no relationship to credit behaviour ,including it would cause data leakage by allowing models to associate ID ranges with default rates specific to the training split.

4.2 Categorical Value Cleaning

Column	Undocumented Values	Count	Action
EDUCATION	0, 5, 6	468 rows (1.6%)	Merged into category 4 ("other")
MARRIAGE	0	54 rows (0.18%)	Merged into category 3 ("other")
SEX	None	0 rows	No action required

4.3 Train / Validation / Test Split

Split	Rows	Default Rate	Purpose
Training (70%)	21,012	22.1%	Model learns patterns from this data
Validation (15%)	4,488	22.1%	Hyperparameter tuning and model comparison
Test (15%)	4,500	22.1%	Final evaluation

Stratified splitting (stratify=y) was applied in both split calls to preserve the 22.1% default rate across all three sets. The consistent default rate across splits was verified after splitting.

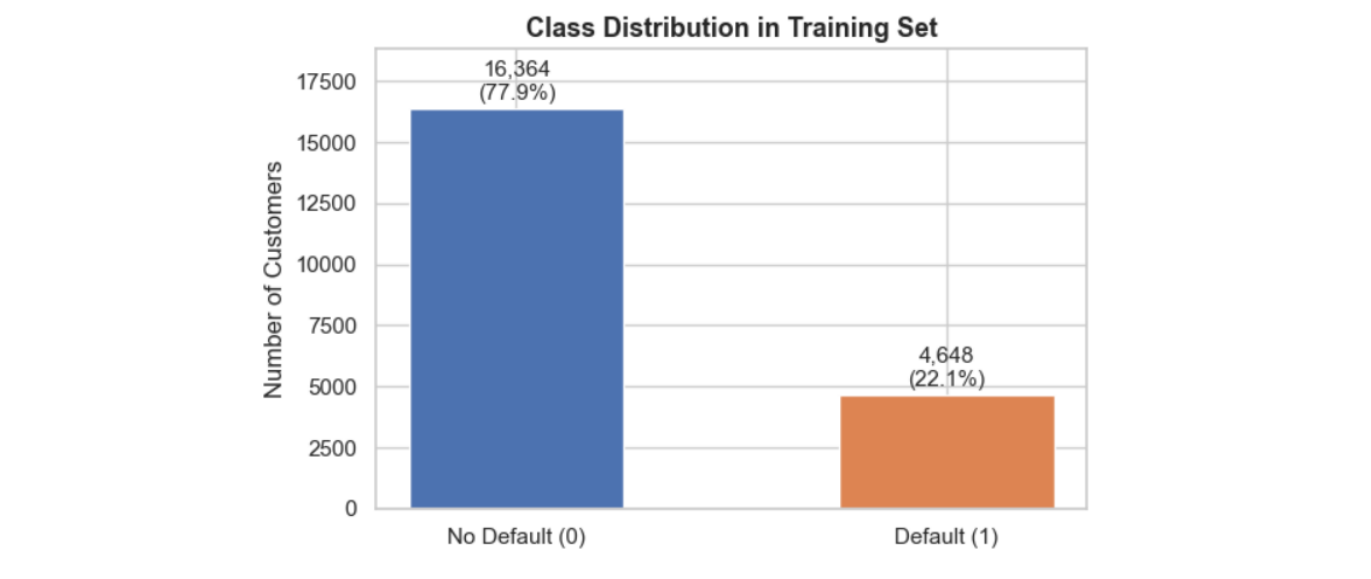
4.4 Feature Scaling

StandardScaler was applied to 20 of 23 features (all columns except SEX, EDUCATION, and MARRIAGE). The three categorical columns were excluded because their integer codes represent categorical labels, not continuous quantities. The scaler was fit exclusively on the training set and then applied (transform only) to validation and test sets. Fitting on the full dataset would constitute data leakage.

Section 5: Exploratory Data Analysis (EDA)

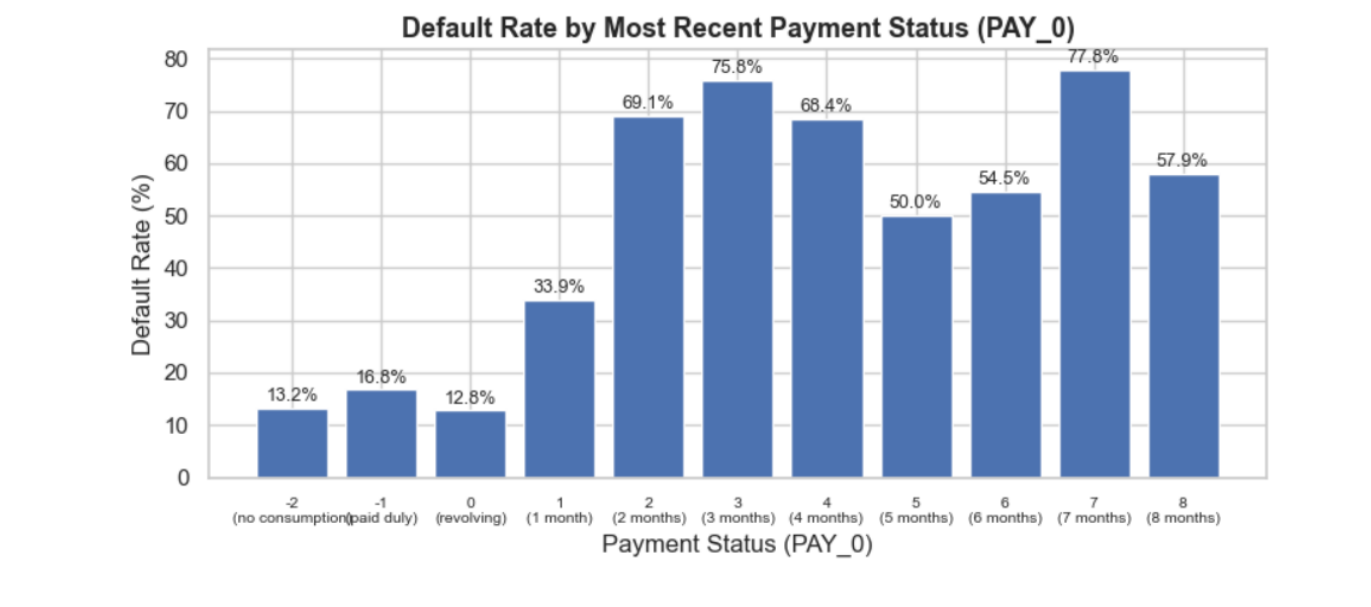
Four visualisations were produced, each chosen to answer a specific question and directly inform modelling decisions. Plots were produced using matplotlib and seaborn with a consistent whitegrid style.

5.1 Plot 1 — Class Distribution



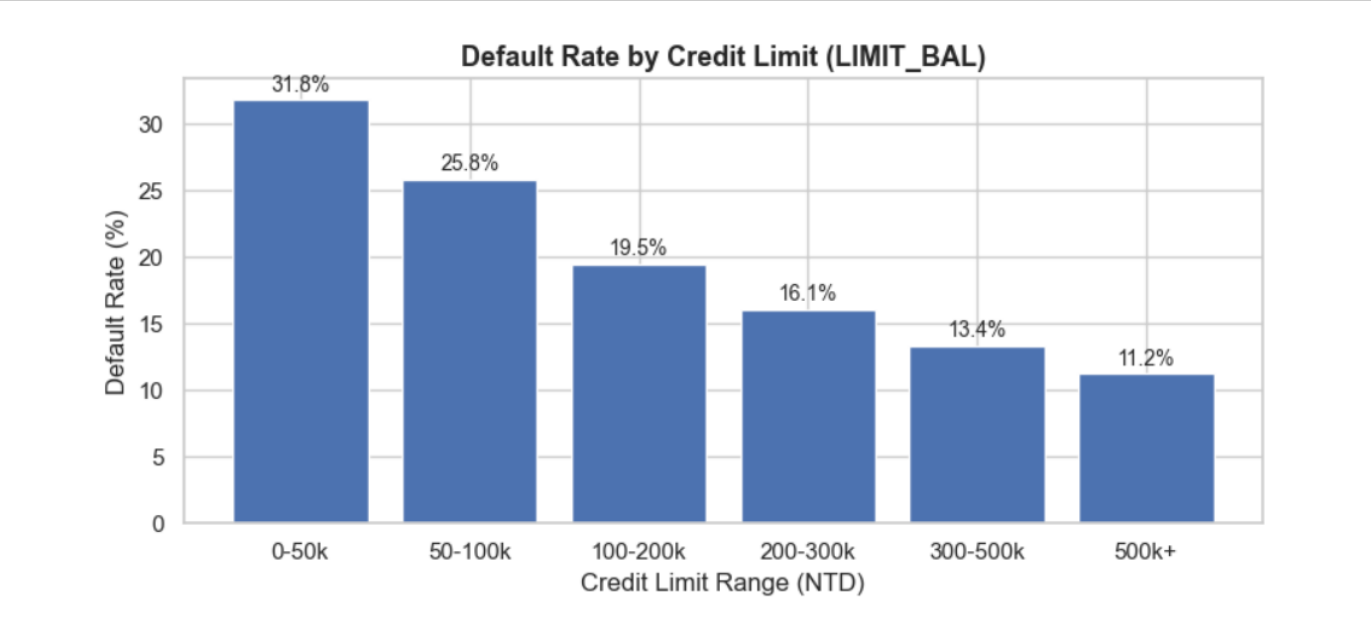
The training set contains 16,364 non-default and 4,648 default cases (22.1% default rate). A model predicting "no default" for every customer would achieve 77.9% accuracy while being completely useless. This finding directly justifies using class_weight="balanced" in all models and prioritising F1-score and ROC-AUC as primary evaluation metrics throughout the study.

5.2 Plot 2 — Default Rate by Payment Status (PAY_0)



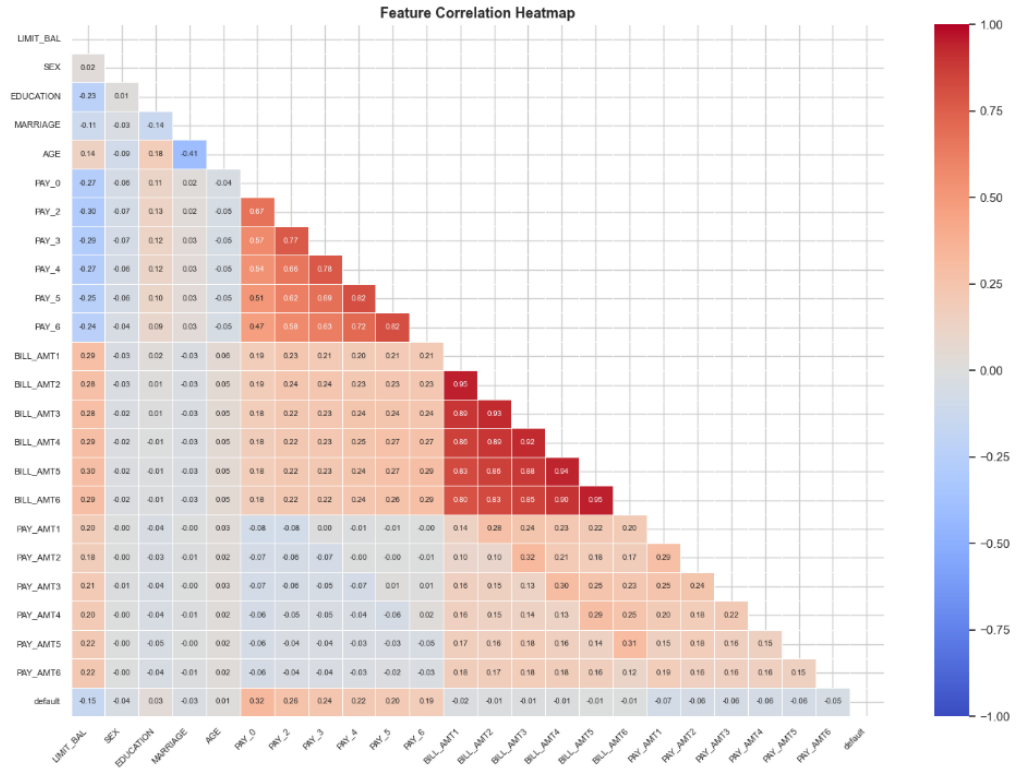
Default rate increases sharply with payment delay. Customers who paid on time (-1) show approximately 10% default rate, while those with two or more month delays exceed 60%. The steep jump between value 0 (revolving credit) and value 1 (one month late) indicates that any missed payment is a strong warning signal. PAY_0 was confirmed as the strongest single predictor in the dataset through both this plot and the correlation heatmap.

5.3 Plot 3 — Default Rate by Credit Limit (LIMIT_BAL)



Customers with credit limits of 0-50k NTD default at approximately 30%, while those with limits above 500k default at under 10%. The monotonically decreasing pattern confirms the hypothesis that credit limit functions as a proxy for the bank's own historical risk assessment. LIMIT_BAL therefore carries both predictive signal about future behaviour and historical signal about the bank's prior evaluation.

5.4 Plot 4 — Feature Correlation Heatmap



The six BILL_AMT columns form a high-correlation block (0.7-0.9), as expected from six consecutive monthly statements for the same account. PAY_0 shows the strongest positive correlation with the target variable, consistent with Plot 2. This multicollinearity will not affect Decision Tree or kNN, but may inflate Logistic Regression coefficient estimates for the BILL_AMT group

Section 6: Modelling

Three classifiers were implemented representing different algorithmic families: a linear model (Logistic Regression), a tree-based model (Decision Tree), and an instance-based model (kNN). This diversity enables a meaningful comparison of decision boundary types, interpretability, and performance on this dataset.

6.1 Logistic Regression

Logistic Regression models the probability of default as a logistic function of a linear combination of input features. It is the standard baseline for binary credit risk classification in industry. Key hyperparameter choices:

- class_weight="balanced" :automatically compensates for the 22% class imbalance by weighting minority-class samples inversely proportional to their frequency.
- max_iter=1000 :default of 100 iterations was insufficient for convergence on this 23-feature dataset.
- Solver: lbfgs (default) : efficient for small-to-medium tabular datasets with L2 regularisation.

6.2 Decision Tree

A Decision Tree partitions the feature space through binary splits, selecting the feature and threshold that maximise Gini purity at each node. An untuned baseline with no depth limit achieved ROC-AUC of 0.622 on the validation set:

Parameter	Values Tested	Best Value	Effect
max_depth	3, 5, 7, 10, 15	5	Limits tree depth; prevents memorisation
min_samples_split	2, 10, 20, 50	50	Node must have 50+ samples to split
min_samples_leaf	1, 5, 10	5	Every leaf must contain 5+ samples

GridSearchCV used 5-fold cross-validation with roc_auc scoring across 60 parameter combinations (300 total fits). Best cross-validation ROC-AUC: 0.7604.

6.3 k-Nearest Neighbours (kNN)

kNN classifies a customer by finding the k most similar training examples and assigning the majority class. It has no class_weight parameter, so imbalance was addressed by tuning k. GridSearchCV was applied across 28 combinations (140 fits):

Parameter	Values Tested	Best Value	Effect
n_neighbors	3, 5, 7, 11, 15, 21, 31	31	Larger neighbourhood smooths majority-class bias
weights	uniform, distance	uniform	Equal weight to all neighbours
metric	euclidean, manhattan	manhattan	L1 distance more robust to financial data outliers

Best cross-validation ROC-AUC: 0.7546. The jump from k=5 to k=31 was the most impactful tuning decision — larger k reduces the majority-class dominance problem inherent in kNN on imbalanced datasets.

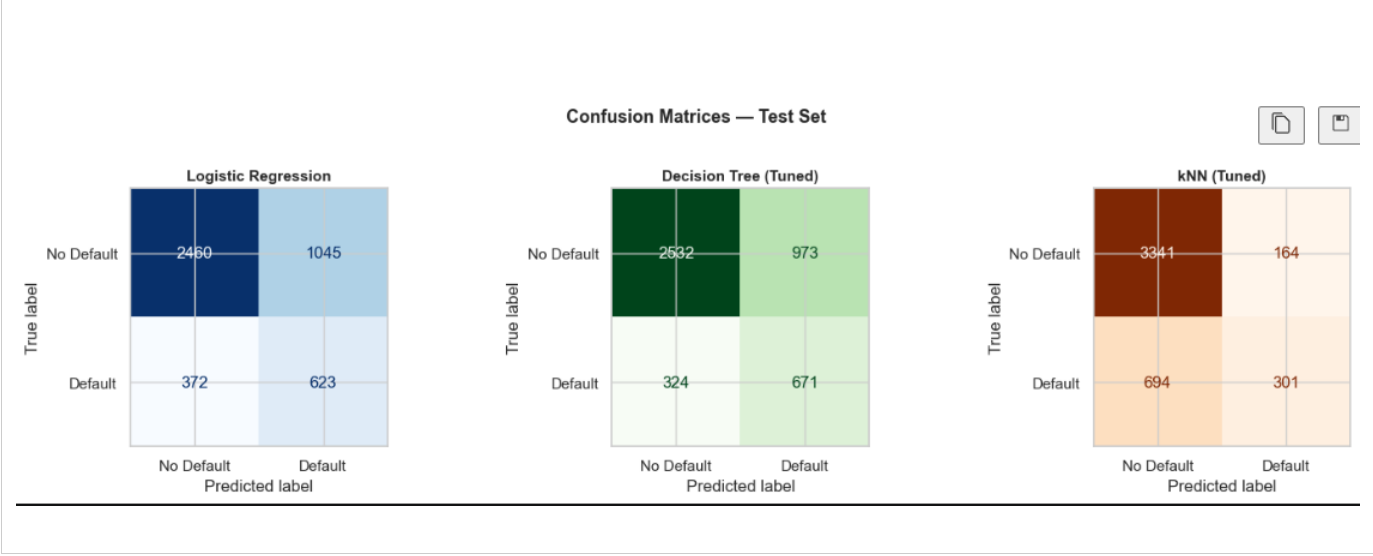
Section 7: Performance Evaluation

All three models were evaluated on the held-out test set (4,500 rows), which was untouched throughout development. These are the final performance numbers for the study.

7.1 Final Test Set Results

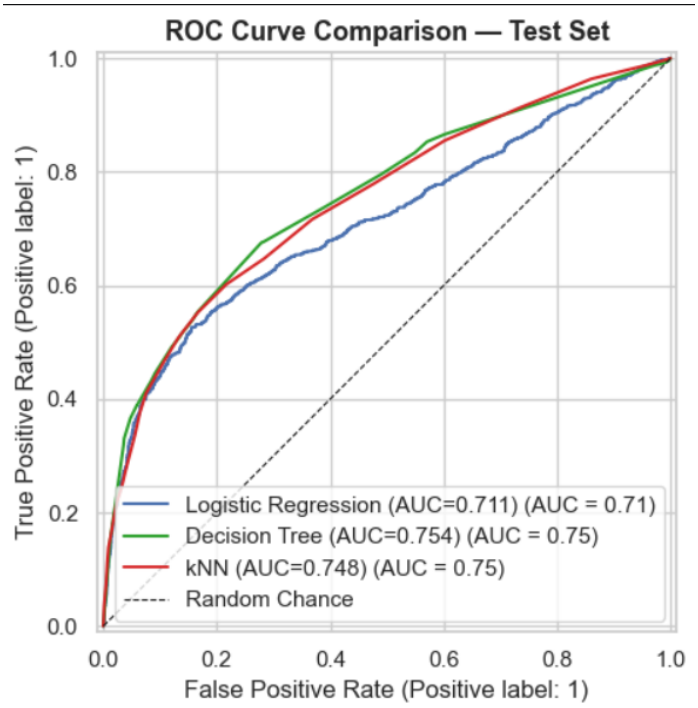
Model	Accuracy	F1-Score	ROC-AUC
Logistic Regression	0.6851	0.4679	0.7107
Decision Tree (Tuned)	0.7118	0.5085	0.7544
kNN (Tuned)	0.8093	0.4123	0.7481

7.2 Confusion Matrices



Model	True Positives	False Negatives	False Positives	True Negatives
Logistic Regression	~623	372	1,045	~2,460
Decision Tree	671	324	973	~2,532
kNN	301	694	164	~3,341

7.3 ROC Curves



The ROC curve plots True Positive Rate against False Positive Rate at every probability threshold. The Decision Tree curve sits above both other models across the full threshold range, confirming its superior discrimination ability. All three models significantly outperform the random-chance diagonal (AUC = 0.5).

Section 8: Model Comparison

8.1 Results Summary

Model	Accuracy	F1-Score	ROC-AUC	FN (missed defaults)	Verdict
Logistic Regression	0.6851	0.4679	0.7107	372	Weakest on all metrics
Decision Tree (Tuned)	0.7118	0.5085	0.7544	324	Best overall — recommended
kNN (Tuned)	0.8093	0.4123	0.7481	694	Misleading accuracy — worst FN

8.2 Why Decision Tree is the Best Model

Decision Tree is the recommended model based on three converging lines of evidence:

- Highest F1-Score (0.5085): F1 penalises both false positives and false negatives. Decision Tree best balances catching defaulters (recall) with avoiding excessive false alarms (precision).
- Highest ROC-AUC (0.7544): The Decision Tree's ROC curve sits above both other models across all probability thresholds, confirming superior overall discrimination ability.
- Fewest missed defaults (FN = 324): The Decision Tree correctly identified 671 of 995 actual defaulters in the test set — more than twice the 301 caught by kNN. In banking, a missed defaulter represents a direct financial loss to the institution.

8.3 Why kNN's 80.9% Accuracy is Misleading

kNN achieved the highest raw accuracy (80.9%) but the lowest F1-Score (0.4123) and the highest false negative count (694). This apparent contradiction is explained by class imbalance. kNN missed 694 out of 995 actual defaulters ,a 70% miss rate on the class that matters. Its high accuracy comes entirely from correctly classifying the 3,500+ non-defaulters, which requires near-zero predictive ability given the 78% majority. This is the classic accuracy paradox in imbalanced classification and is the reason F1-score and ROC-AUC are the appropriate primary metrics for this problem.

8.4 Why Logistic Regression Underperformed

Logistic Regression produced the weakest results on all three metrics. The most likely cause is its linear decision boundary :credit default behaviour involves complex interactions between features that a linear model cannot capture without explicit feature engineering. Additionally, the high multicollinearity among the six BILL_AMT columns (0.7-0.9, identified in EDA) inflates the variance of corresponding coefficients, reducing the model's reliability.

Section 9: Error Analysis

9.1 Misclassification Summary

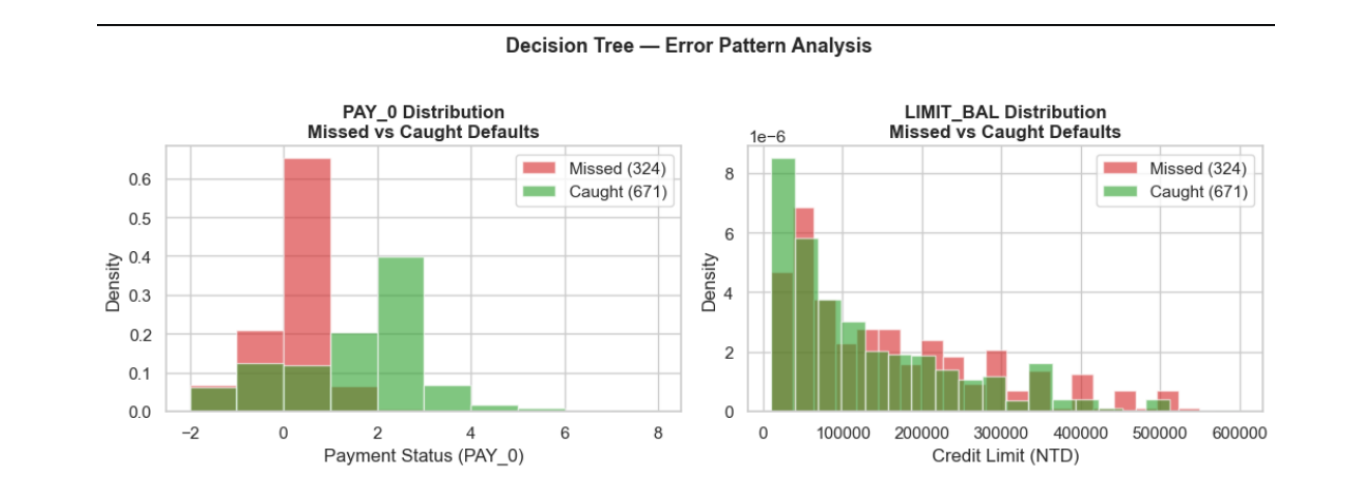
Model	Total Errors	Error Rate	False Negatives	False Positives
Logistic Regression	1,417	31.5%	372 (missed defaults)	1,045 (false alarms)
Decision Tree (Tuned)	1,297	28.8%	324 (missed defaults)	973 (false alarms)
kNN (Tuned)	858	19.1%	694 (missed defaults)	164 (false alarms)

The most important column is False Negatives. In a banking context, a false negative means a customer who will default is predicted as safe because the bank extends credit and incurs the full loss. kNN's low total error count is deeply misleading: 694 of its 858 errors are missed defaults, meaning it fails on the most costly error type at nearly twice the rate of Decision Tree.

9.2 Who Does the Decision Tree Miss?

Analysis of the 324 missed defaults versus the 671 correctly identified defaults for the Decision Tree reveals a clear pattern:

Feature	Missed Defaults (avg)	Caught Defaults (avg)	Interpretation
PAY_0	-0.3	1.1	Missed customers paid on time recently — appear safe
LIMIT_BAL	160,124 NTD	127,943 NTD	Missed customers have higher credit limits
BILL_AMT1	59,682 NTD	45,897 NTD	Missed customers carry higher recent balances
PAY_AMT1	5,015 NTD	2,460 NTD	Missed customers made larger recent payments
AGE	36.5 years	36.0 years	No meaningful difference by age



9.3 Error Pattern Interpretation

The Decision Tree misses defaulters who appear financially healthy in the most recent month: they have low PAY_0 values (recent on-time payments), higher credit limits (suggesting the bank previously trusted them), and were making larger payments recently.

This failure pattern is consistent with a fundamental dataset limitation: the six-month window captures historical behaviour but cannot anticipate future disruptions.

9.4 Limitations

- Six-month window only: Longer payment history would enable detection of gradual financial deterioration patterns invisible in short windows.
- Class imbalance ceiling: Even with class_weight="balanced", the 22% minority class limits how much the models can learn about default patterns.
- kNN production impracticality: kNN requires computing distances to all 21,012 training examples at prediction time. This is computationally impractical for a real bank processing millions of customers.
- BILL_AMT multicollinearity: The high correlation among the six bill amount columns limits Logistic Regression coefficient interpretability.

Section 10: AI Tool Usage & Reflection

AI tools were used primarily to clarify unfamiliar concepts and debug specific errors. All design decisions, model choices, and interpretations were made independently by the team. Every AI explanation was verified by running code and inspecting outputs.

Example 1 — Understanding GridSearchCV

Prompt: "What is GridSearchCV and how does it work?"

The explanation of systematic hyperparameter search via cross-validation helped the team choose meaningful parameter ranges for Decision Tree tuning (e.g., max_depth values of 3, 5, 7, 10, 15 based on understanding of overfitting). The explanation was verified by observing the baseline-to-tuned AUC improvement (0.622 → 0.754), which confirmed the intuition about depth constraints.

What was wrong or misleading: Nothing conceptually incorrect, but the explanation alone was insufficient because the team had to select parameter ranges and interpret results independently.

Example 2 — Debugging a Code Error

Error encountered: "ValueError: Found input variables with inconsistent numbers of samples: [25500, 30000]"

The bug was identified as using stratify=y (30,000 rows) in the second train_test_split call instead of stratify=y_temp (25,500 rows). The fix was a single parameter change. The fix was confirmed by checking split sizes: 21,012 / 4,488 / 4,500 summing to 30,000.

Example 3 — Clarifying Metric Interpretation

Prompt: "kNN has the highest accuracy but lowest F1. Why is Decision Tree considered better here?"

The clarification of the accuracy paradox in imbalanced classification helped the team articulate why kNN's 80.9% accuracy is misleading. The explanation was verified directly from confusion matrix results: kNN's False Negative count of 694 versus Decision Tree's 324 confirmed that kNN fails on the most important error type despite its high accuracy.

Conclusion

This study implemented and evaluated three supervised classification models on the Credit Card Default dataset (30,000 instances, 23 features, binary target). The preprocessing pipeline delivered a clean, correctly split, and scaled dataset with all design decisions explicitly justified. EDA revealed a 22.1% class imbalance, identified PAY_0 as the strongest predictor, confirmed LIMIT_BAL as a meaningful feature, and flagged multicollinearity among the BILL_AMT columns.

Among the three classifiers, the tuned Decision Tree (max_depth=5, min_samples_split=50, min_samples_leaf=5) is the recommended model, achieving the highest F1-Score (0.5085) and ROC-AUC (0.7544) on the held-out test set. It correctly identified 671 of 995 actual defaulters which is more than twice the 301 caught by kNN. Logistic Regression provided a meaningful linear baseline but was limited by its inability to capture non-linear feature interactions. kNN achieved the highest raw accuracy (80.9%) but the worst performance on the metrics that matter for imbalanced binary classification.

Error analysis revealed that all models struggle with a specific subgroup: customers who appear financially healthy in the most recent month (PAY_0 near -1, higher credit limits, recent on-time payments) but default shortly after. This pattern requires longitudinal data beyond the six-month window available in this dataset and represents the primary opportunity for future improvement.